

# PRADEEP KUMAR

AI PRODUCT ENGINEER · FORWARD-DEPLOYED ENGINEER · AGENTIC SYSTEMS

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## PERSONAL PROJECT

### Crucible — Self-hosted VS Code coding-agent platform

2026

crucible-ai.pages.dev | GitHub | Python, TypeScript, Rust | 14 releases | Apache 2.0

- **Safe agent execution.** Built a FastAPI ReAct orchestrator that performs work in isolated shadow workspaces, requires explicit plan/edit/command/MCP approval, and promotes only reviewed file changes to the user's working tree.
- **Code intelligence and context engineering.** Built a Rust indexer using tree-sitter and live LSP resolution across six language families, exposed as an interactive dependency explorer; added cross-session hybrid memory with sqlite-vec, BM25, reranking, and anchored summaries.
- **Shippable developer product.** Released a self-provisioning VS Code extension and cross-platform runtime that checksum-verifies nine bundled developer tools; supports ten model transports, including local and watsonx providers, with graceful capability degradation and release automation.

## EXPERIENCE

### Lead Engineer IBM

Feb 2024 – Present  
Bengaluru, India

- **Platform architecture.** Promoted from Senior AI Engineer to **Lead Engineer**; led production rollout of Cloud Pak for Data, IBM Knowledge Catalog (now watsonx.data intelligence), watsonx Orchestrate, and watsonx.governance on customer-managed OpenShift for Rogers. Defined high-availability topology, cluster sizing, hardware specifications, prerequisites, and runbooks; used oc and cpd-cli to resolve operator reconciliation failures and deliver a stable, production-ready platform for governed AI and data workloads.
- **Customer AI systems.** Architected and deployed 3+ enterprise-grade RAG systems on watsonx.ai and watsonx.orchestrate across finance, HR, and energy domains, serving **5,000+** internal and client users; built the supporting inference, ranking, and feedback microservices, containerized with Podman on IBM Cloud Code Engine with auto-scaling, absorbing 3x traffic growth without redesign.
- **IBM Client Engineering asset.** Built **Ask-Git**, a production code-intelligence accelerator adopted by IBM Client Engineering through CE Coach and reused across client engagements; combined LangChain, GitHub APIs, and vector stores to query commits, file-level changes, and pull-request metadata across repositories.
- **Retrieval quality.** Designed end-to-end pipelines (ingestion, chunking, embedding, vector indexing, ranking, LLM synthesis) processing **1M+ documents at sub-2s** median latency; implemented hybrid search across Elasticsearch and Watson Discovery, improving retrieval precision by 28% and reducing hallucination-related escalations by 35%.
- **Reliable agents.** Designed structured JSON contract outputs and observability for tool-calling pipelines, reducing parsing failures by **90%** and debugging time by 45%; drove evaluation-driven prompt engineering, LoRA adapter fine-tuning, and grounding strategies, improving answer consistency by 22% across evaluation benchmarks.
- **Insurance-agent product.** Built a **LangGraph Call Transcript Analyzer** for CI Financial, enabling insurance agents to replace manual full-transcript skims with structured call summaries, tone, action items, resolved issues and risks, and sales opportunities to inform policy conversations and selling.
- **Client-to-platform reuse.** Built a **LangGraph multi-agent resume-to-JD matching workflow** for OPG, cutting manual review effort by **40%**; productized the client-proven use case as a reusable watsonx Orchestrate asset and extended the portfolio with natural-language-to-SQL and vision-language document-extraction agents.
- **Commercial and organizational leverage.** Influenced **USD 534K** in customer revenue (USD 200K annual recurring) across three engagements; drove watsonx Orchestrate adoption across APAC through enablement, reusable assets, mentoring, and architecture reviews.

## EARLIER EXPERIENCE

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### AI Engineer → Senior AI Engineer Inflect

Sep 2021 – Feb 2024  
Bengaluru, India

- Led six end-to-end AI deployments for global FMCG retail analytics, spanning Share-of-Shelf, On-Shelf Availability, image retrieval, segmentation, and cross-modal LLM services.
- Built product-identification models that computed Share-of-Shelf and On-Shelf Availability for major brands; implemented reusable detection and OCR components that standardized capabilities across client deployments.
- Built a cross-modal BLIP-2 brand-identification service that eliminated periodic product annotation, pairing text and image understanding with reusable client-facing AI components.
- Improved inference efficiency by 25% with Patch2Pix and SAM pipelines; DINO-based retrieval reduced manual workflows by 35%, while LLM services cut annotation effort by 40%.
- Architected deep-learning microservices with Docker, Kubernetes, Redis, nginx, and TensorRT.

### Computer Vision Engineer Drishte

Jan 2020 – Sep 2021  
Bengaluru, India

- Built crowd-analytics and mobile face-recognition systems covering tracking, detection, classification, and re-identification.
- Deployed models to Jetson Nano and edge devices with TensorRT, OpenVINO, and TensorFlow Lite; improved accuracy by 20% and automated CI/CD to cut deployment time by 30%.

## TOOLKIT

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**Agentic AI:** LangGraph, LangChain, Model Context Protocol (MCP), ReAct, multi-agent orchestration, tool calling, structured outputs, JSON Schema, evaluation-driven prompting, LoRA, VLMs

**Retrieval & Data:** RAG, hybrid search, Elasticsearch, Watson Discovery, FAISS, sqlite-vec, BM25, cross-encoder reranking, chunking/embedding pipelines, natural-language-to-SQL

**Platform & Delivery:** Python, TypeScript, Rust, SQL, FastAPI, OpenShift (oc, cpd-cli), Kubernetes, Docker, Podman, Kafka, Redis, PostgreSQL, Terraform, Vault, GitHub Actions, AWS, Azure, GCP

**AI Platforms:** watsonx.orchestrate (ADK), watsonx.ai, watsonx.governance, IBM Knowledge Catalog, Cloud Pak for Data, OpenAI, Anthropic, Gemini, vLLM, Ollama, NVIDIA NIM

**Computer Vision:** PyTorch, TensorFlow, OpenCV, TensorRT, OpenVINO, SAM, BLIP-2, DINO

## EDUCATION

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BE, Electronics and Communication, SLN College, VTU, Karnataka

2016 – 2020 · CGPA: 8.0/10